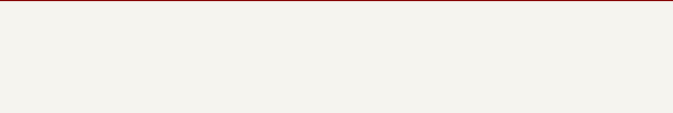


OSBAMS Rev.2 Controller – RELEASE CANDIDATE 1 (review only; NOT FOR FABRICATION)

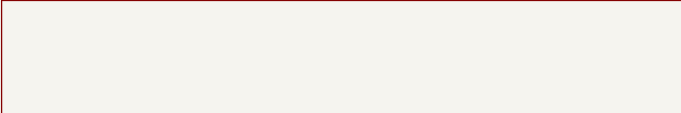
Battery discharge current never enters this PCB. 44 V / 10 A MAX. Hierarchical sheets use global labels for cross-sheet nets.

01_Power_Input



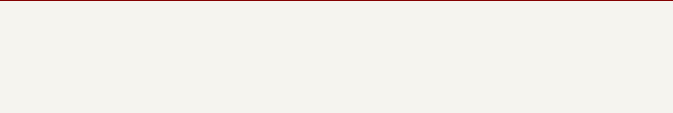
File: 01_Power_Input.kicad_sch

02_Buck_3V3



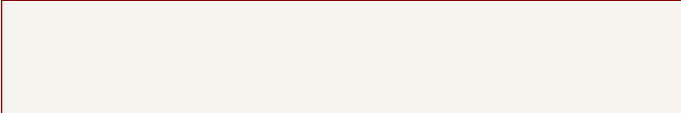
File: 02_Buck_3V3.kicad_sch

03_MCU



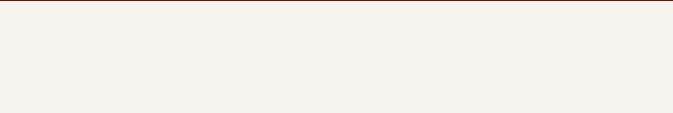
File: 03_MCU.kicad_sch

04_Host_Isolated



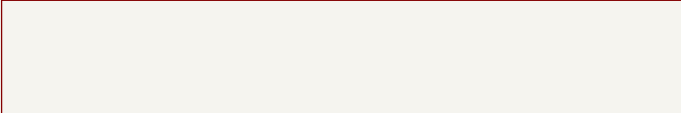
File: 04_Host_Isolated.kicad_sch

05_INA228_Shunt



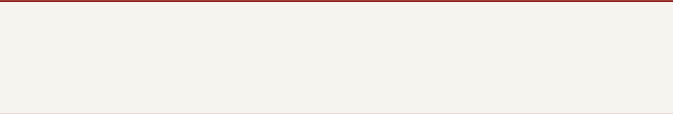
File: 05_INA228_Shunt.kicad_sch

06_ADC_PackSense



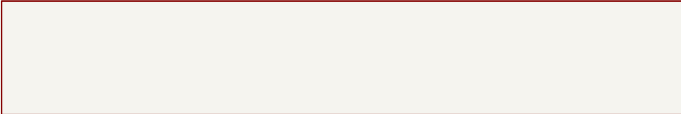
File: 06_ADC_PackSense.kicad_sch

07_Coil_Driver



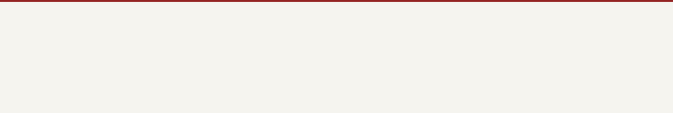
File: 07_Coil_Driver.kicad_sch

08_Status_Sense



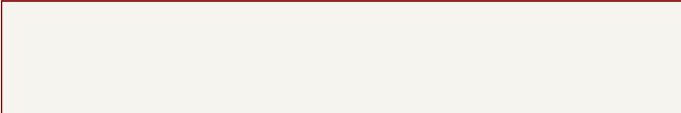
File: 08_Status_Sense.kicad_sch

09_Temp_Probe



File: 09_Temp_Probe.kicad_sch

10_TestPoints_Mech



File: 10_TestPoints_Mech.kicad_sch

RELEASE CANDIDATE 1 – NOT FOR FABRICATION

OSBAMS

Sheet: /
File: OSBAMS_Rev2_RC1.kicad_sch

Title: OSBAMS Rev.2 Controller

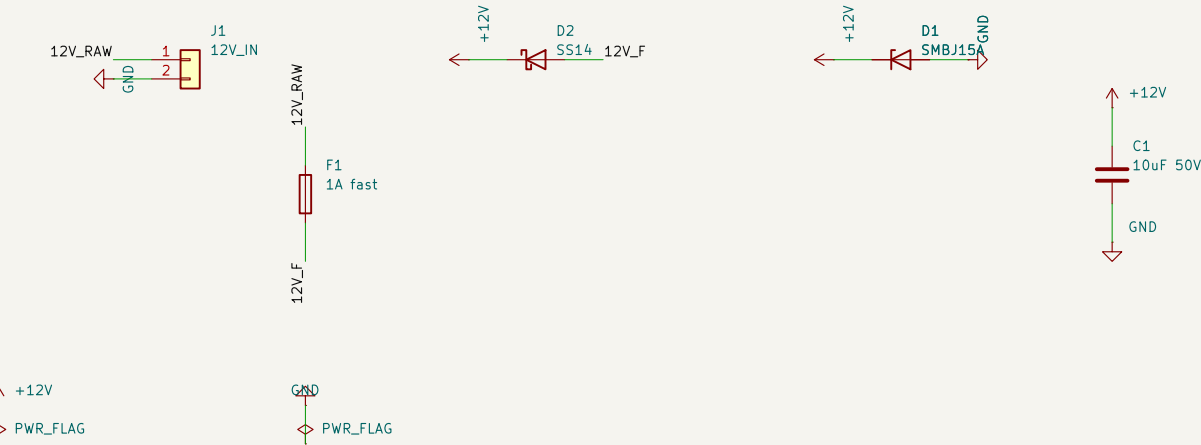
Size: A3	Date: 2026-10-01	Rev: RC1
KiCad E.D.A. 10.0.6		Id: 1/11

Power input and protection

OSBAMS Rev.2 Controller RC1 – REVIEW CANDIDATE, NOT FOR FABRICATION. Evidence: VERIFIED_LOCAL / USER_RELAYED_MANUFACTURER / UNVERIFIED (see evidence register).

J1: 12 V control input (XDR–75–12 set and verified at 12.0 V). F1 1 A fast, D2 Schottky reverse–polarity block (0.4 V drop), TVS D1 SMBJ15A across +12V AFTER D2 (reverse polarity is blocked, not clamped).

Diode polarity: Device:D / D_Schottky / D_Zener pin 1 = cathode (K); KiCad SMA/SMB footprints pad 1 = cathode – verified by tools in checks.py (netlist polarity check).



Manufacturer–document verification is a fabrication gate
RELEASE CANDIDATE 1 – NOT FOR FABRICATION
OSBAMS Rev.2 Controller
Sheet: /01_Power_Input/
File: 01_Power_Input.kicad_sch

Title: Power input and protection

Size: A4	Date: 2026–10–01	Rev: RC1
KiCad E.D.A. 10.0.6		Id: 2/11

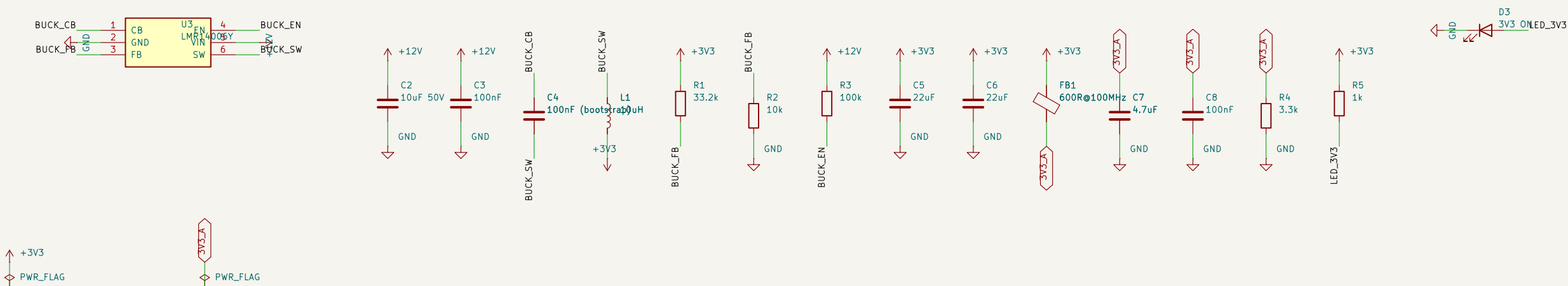
12 V $\rightarrow$ 3.3 V buck, filtered 3V3_A

OSBAMS Rev.2 Controller RC1 – REVIEW CANDIDATE, NOT FOR FABRICATION. Evidence: VERIFIED_LOCAL / USER_RELAYED_MANUFACTURER / UNVERIFIED (see evidence register).

LMR14006Y (4–40 V, 600 mA, 2.1 MHz, VREF 0.765 V: user-relayed). $V_{out} = 0.765 \times (1 + R1/R2) = 0.765 \times (1 + 33.2k/10k) = 3.305 \text{ V}$.

Hot loop: C2/C3 – U3 VIN/GND – SW – L1 kept tiny in layout. No 5 V rail. 3V3_A = 3V3 through ferrite FB1 + 4.7 uF + 100 nF (VDDA, INA228).

R4 3.3k: permanent bleeder on 3V3_A; limits the rail to $< 1\text{ V}$ if PACK+ back-feeds the MCU pin while the board is unpowered (calculations section 1.1).



Manufacturer-document verification is a fabrication gate
RELEASE CANDIDATE 1 – NOT FOR FABRICATION

OSBAMS Rev.2 Controller

Sheet: /02_Buck_3V3/
File: 02_Buck_3V3.kicad_sch

Title: 12 V $\rightarrow$ 3.3 V buck, filtered 3V3_A

Size: A3	Date: 2026-10-01
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Rev: RC1

Size: A5	
KiCad E.D.A. 10.0.6	

Id: 3/11

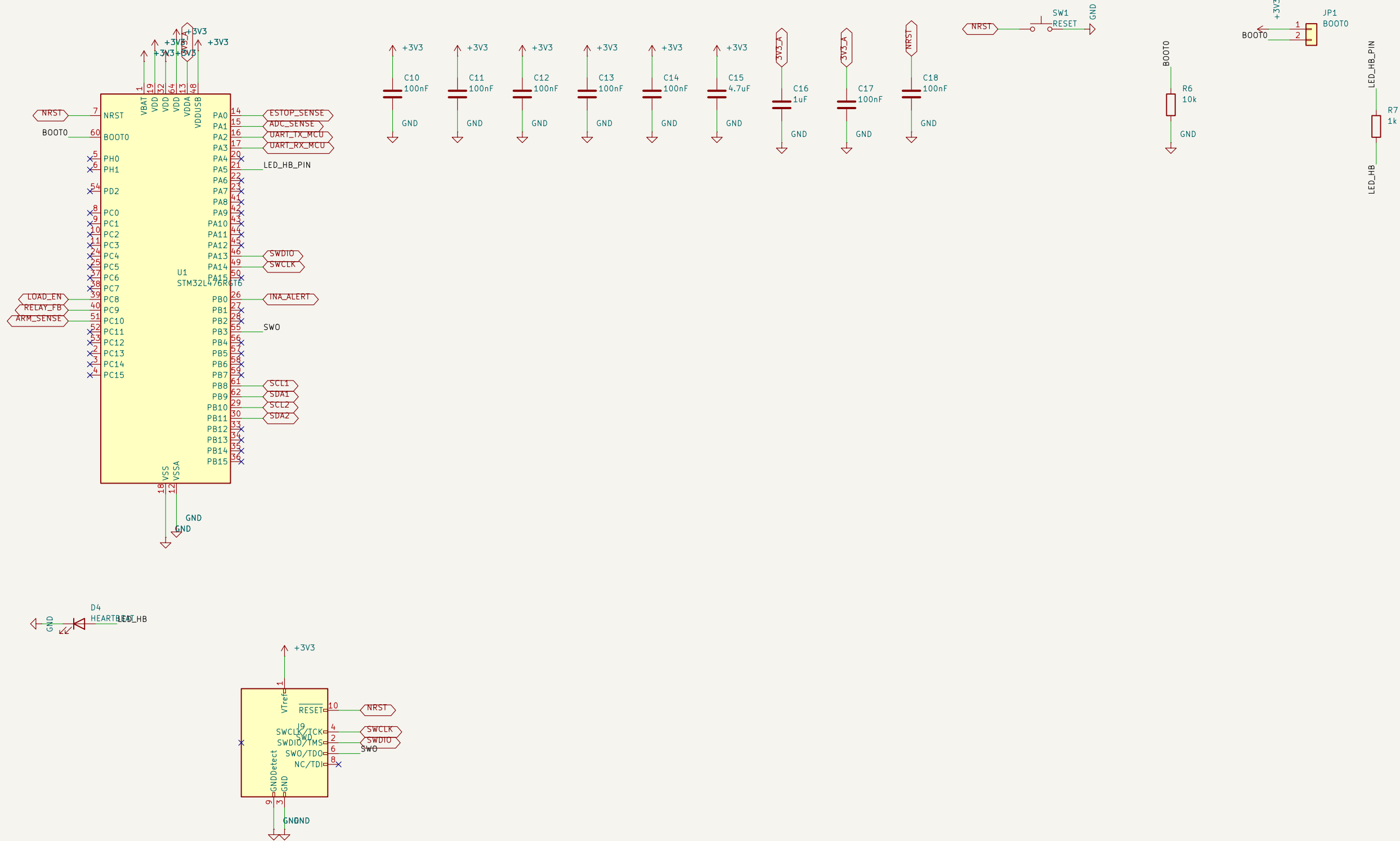
STM32L476RGT6, reset/boot/SWD

USBAMS Rev.2 Controller RC1 — REVIEW CANDIDATE, NOT FOR FABRICATION. Evidence: VERIFIED_LOCAL / USER_RELAYED_MANUFACTURER / UNVERIFIED (see evidence register).

STM32L476RGT6 direct. HSI16 -> PLL 80 MHz (no crystal), no LSE, USB unused. PA0 ESTOP_SENSE (LOW = healthy), PC9 RELAY_FB (active low), PC10 ARM_SENSE (HIGH = armed, status only).

PA1 ADC_SENSE (VREFINT-corrected), PA2/PA3 USART2, PB8/PB9 I2C1 (INA228), PB10/PB11 I2C2 (TC74, <= 100 kHz), PB0 INA228 ALERT, PC8 LOAD_EN (default low), PA5 heartbeat, SWD on J9.

Unused pins carry no-connect flags and are configured analog/pulled low in firmware. BOOT0: 10k pull-down + jumper JP1 to 3V3 (ROM bootloader).



Manufacturer–document verification is a fabrication gate
RELEASE CANDIDATE 1 – NOT FOR FABRICATION

OSBAMS Rev.2 Controller

Sheet: /03_MCU/
File: 03_MCU.kicad_sch

Title: STM32L476RGT6, reset/boot/SWD

Size: A3	Date: 2026-10-01
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KiCad E.D.A. 10.0.6

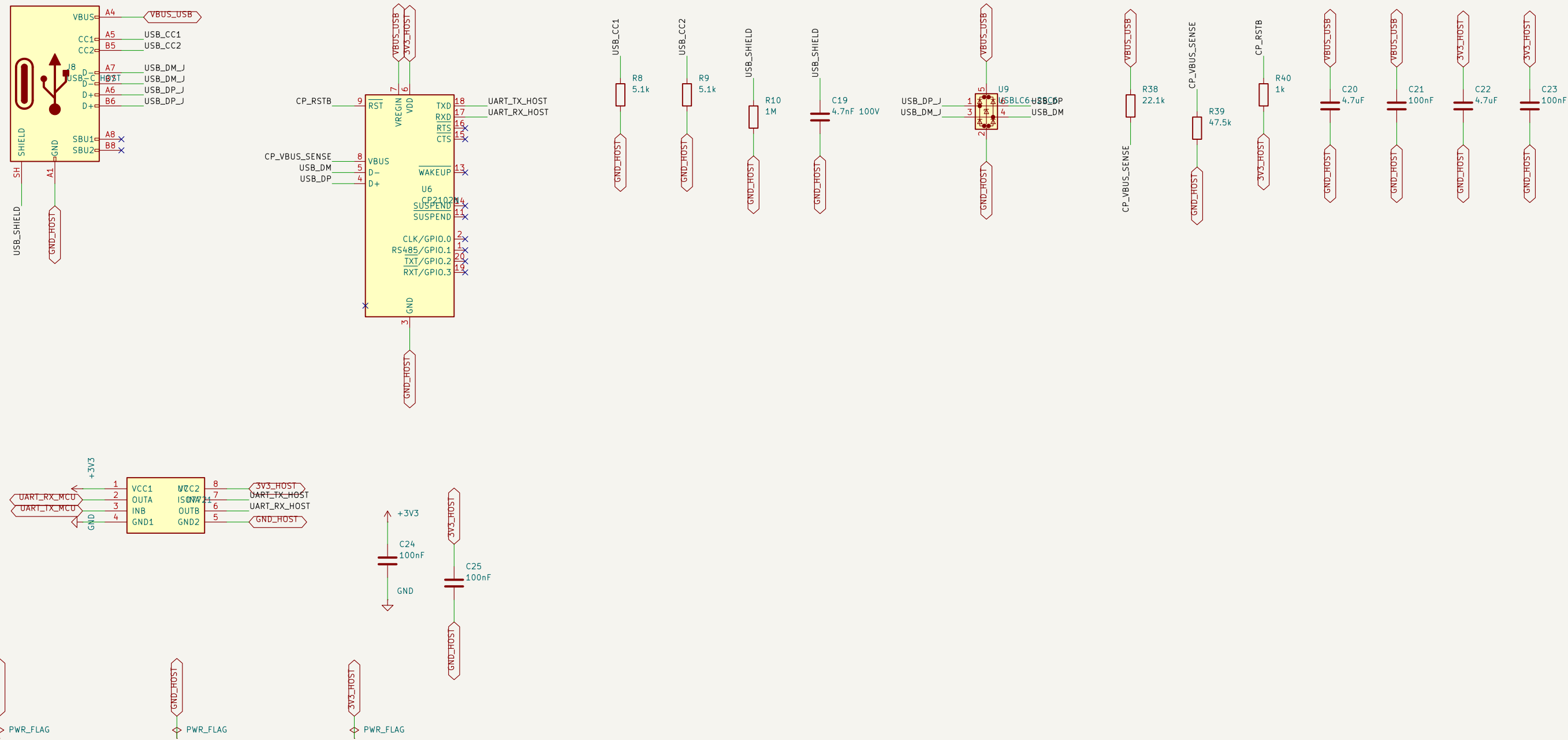
Rev: RC1

Id: 4/11

USART2 -> ISO7721 -> CP2102N -> USB-C

OSBAMS Rev.2 Controller RC1 – REVIEW CANDIDATE, NOT FOR FABRICATION. Evidence: VERIFIED_LOCAL / USER_RELAYED_MANUFACTURER / UNVERIFIED (see evidence register).

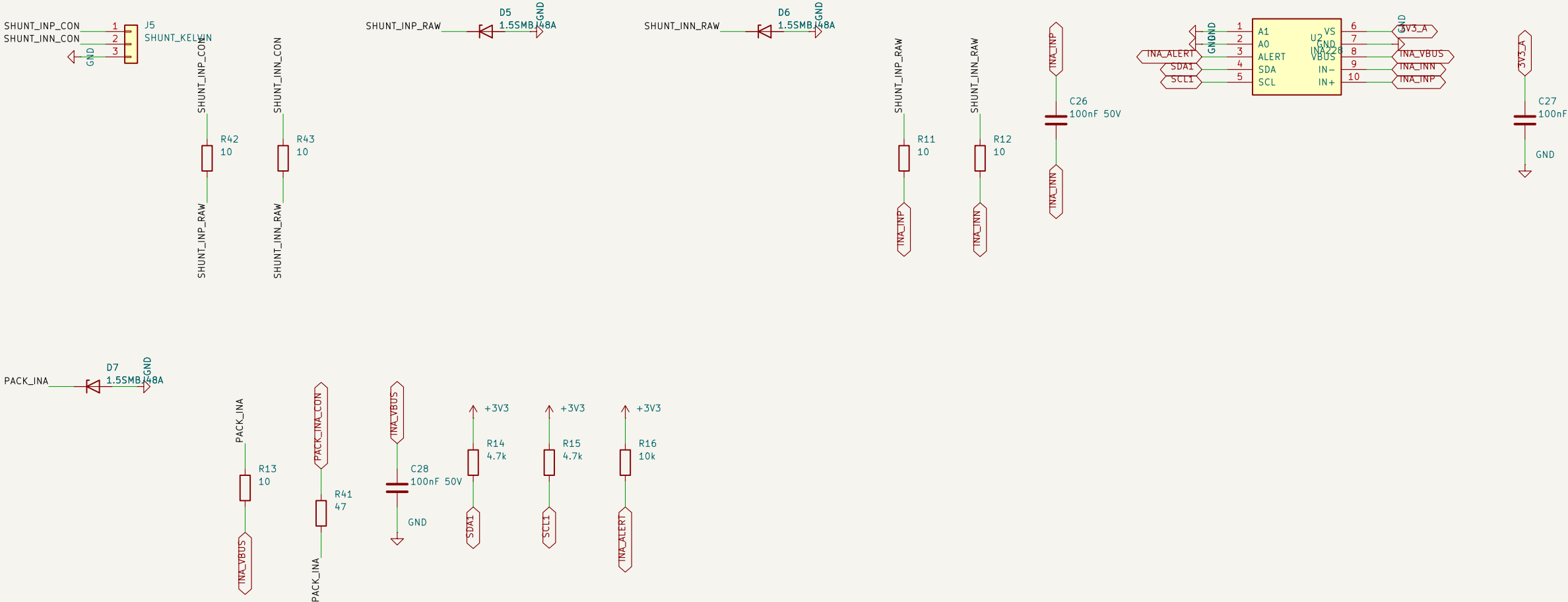
USART2 -> ISO7721 -> CP2102N -> USB-C -> Raspberry Pi 5. ISO7721 has NO isolated power: VCC1 = +3V3 (controller side), VCC2 = 3V3_HOST from the CP2102N regulator (VREGIN from USB VBUS).
GND (controller) and GND_HOST (USB side) are separate nets: they are never joined. USB shield: 1 M || 4.7 nF to GND_HOST. CC1/CC2 5.1 k pull-downs. All values UNVERIFIED against the CP2102N reference design.



INA228 + shunt Kelvin interface

OSBAMS Rev.2 Controller RC1 – REVIEW CANDIDATE, NOT FOR FABRICATION. Evidence: VERIFIED_LOCAL / USER_RELAYED_MANUFACTURER / UNVERIFIED (see evidence register).

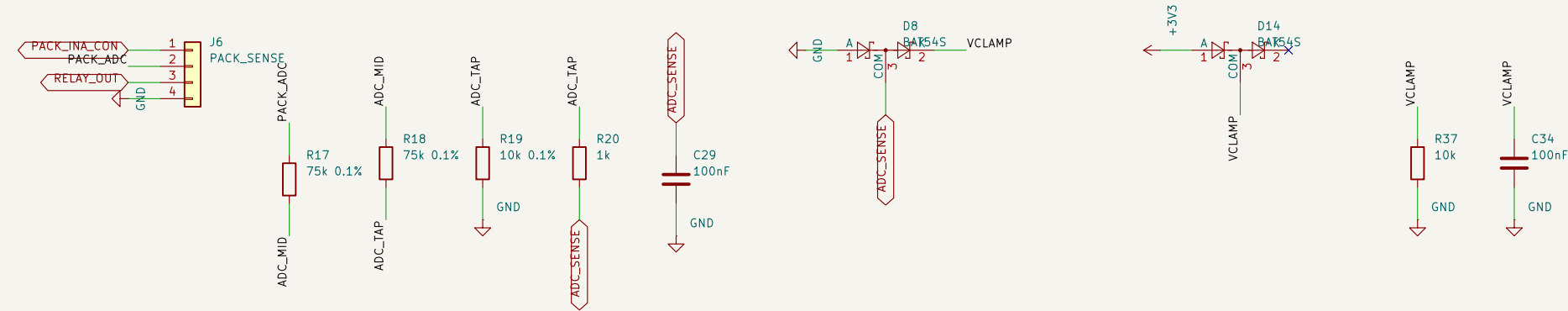
INA228 reads the external Bourns RSA-20-50 (2.5 mOhm, 20 A / 50 mV) through J5 (Kelvin). ADCRANGE = 0 (+/-163.84 mV); SHUNT_CAL = 1250 at IMAX = 20 A (firmware change pending).
IN+/IN- filter 2 x 10 ohm + 100 nF; TVS 1.5SMBJ48A (77.4 V clamp vs 85 V abs max) on each pack-level input. VBUS comes from PACK_INA (upstream of K1) on its own lead. A0 = A1 = GND -> address 0x40.
INA228 pin map is an INA226-family ANALOGY and is UNVERIFIED: verify against the TI datasheet before fabrication.



Independent ADC channel and pack-sense connector

OSBAMS Rev.2 Controller RC1 – REVIEW CANDIDATE, NOT FOR FABRICATION. Evidence: VERIFIED_LOCAL / USER_RELAYED_MANUFACTURER / UNVERIFIED (see evidence register).

PACK_ADC -> 75k -> 75k -> ADC_TAP -> 10k -> GND (16:1; 44 V -> 2.750 V); 1 k + 100 nF; BAT54S clamp: lower diode to GND, upper diode to the isolated rail VCLAMP (fed from 3V3 through D14, bled by R37) -> PACK+ cannot back-power 3V3.
J6 pin 4 (GND_SENSE) is the single bond between controller GND and pack-negative. Separate PACK_ADC lead from the INA228 VBUS lead.



Manufacturer-document verification is a fabrication gate
RELEASE CANDIDATE 1 – NOT FOR FABRICATION
OSBAMS Rev.2 Controller
Sheet: /06_ADC_PackSense/
File: 06_ADC_PackSense.kicad_sch

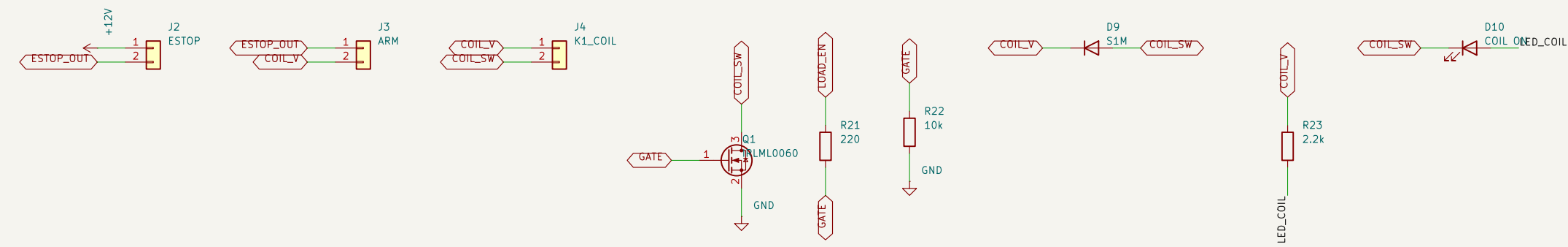
Title: Independent ADC channel and pack-sense connector

Size: A3	Date: 2026-10-01	Rev: RC1
KiCad E.D.A. 10.0.6		Id: 7/11

E-stop / ARM / K1 coil chain and MOSFET driver

OSBAMS Rev.2 Controller RC1 – REVIEW CANDIDATE, NOT FOR FABRICATION. Evidence: VERIFIED_LOCAL / USER_RELAYED_MANUFACTURER / UNVERIFIED (see evidence register).

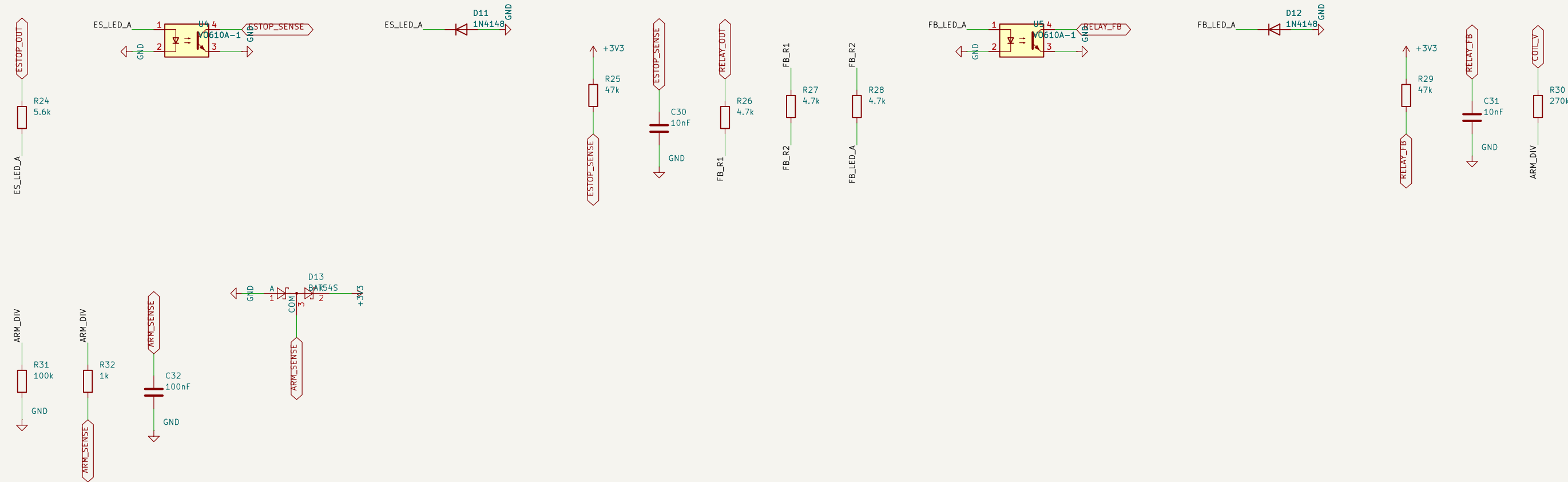
Coil chain: +12V -> J2 E-stop (Eaton M22-PV-K02 NC) -> J3 ARM (C&K T102SHZQE) -> J4 K1 coil -> Q1 -> GND. E-stop and ARM physically open the coil supply; firmware cannot bypass either.
Q1 gate: 220 ohm series, 10k pull-down (default OFF). Flyback D9 cathode on the +12 V side. No fast-release TVS in RC1. Battery discharge current never flows through this board.



E-stop, ARM and relay-feedback sensing

OSBAMS Rev.2 Controller RC1 - REVIEW CANDIDATE, NOT FOR FABRICATION. Evidence: VERIFIED_LOCAL / USER_RELAYED_MANUFACTURER / UNVERIFIED (see evidence register).

ESTOP_SENSE: ESTOP_OUT -> 5.6k -> V0610A LED: collector pull-up 47k -> PA0 (LOW = healthy; open/broken = HIGH). RELAY_FB: K1 load side -> 3 x 4.7k (1206) -> V0610A LED: pull-up 47k -> PC9 (active low).
Resistors sized from the user-relayed V0610A-1 CTR guarantee (min 13 % @ 1 mA, 40 % @ 10 mA) with temperature/aging derates and 2x saturation margin: supported 30-44 V pack range checked down to 24 V (calculations section 4).
ARM_SENSE: 270k/100k divider, 1k, 100 nF, BAT54S clamp - sensing only, never permission to energize.



Manufacturer-document verification is a fabrication gate
RELEASE CANDIDATE 1 - NOT FOR FABRICATION

OSBAMS Rev.2 Controller

Sheet: /08_Status_Sense/

File: 08_Status_Sense.kicad_sch

Title: E-stop, ARM and relay-feedback sensing

Size: A2 Date: 2026-10-01

Rev: RC1

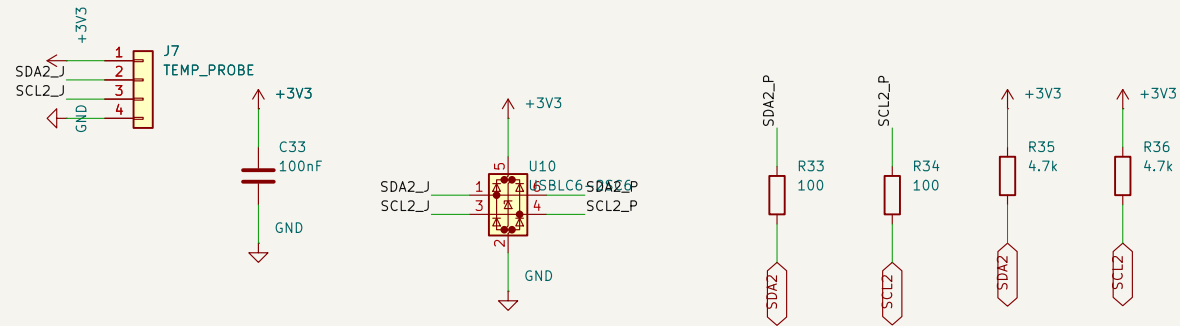
KiCad E.D.A. 10.0.6

Id: 9/11

Remote TC74 probe interface (I2C2)

OSBAMS Rev.2 Controller RC1 – REVIEW CANDIDATE, NOT FOR FABRICATION. Evidence: VERIFIED_LOCAL / USER_RELAYED_MANUFACTURER / UNVERIFIED (see evidence register).

Remote TC74A5-3.3VAT probe on I2C2 (PB10/PB11), ≤ 100 kHz, cable ≤ 1.5 m. Probe has its own 100 nF at the sensor. 4.7k pull-ups fitted (2.2k alternative), 100 ohm series, 2-channel ESD.



Manufacturer–document verification is a fabrication gate

RELEASE CANDIDATE 1 - NOT FOR FABRICATION

OSBAMS Rev.2 Controller

Sheet: /09_Temp_Probe/

File: 09_Temp_Probe.kicad_sch

Title: Remote TC74 probe interface (I2C2)

Size: A4	Date: 2026-10-01
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Size: A4	
KiCad E.D.A. 10.0.6	

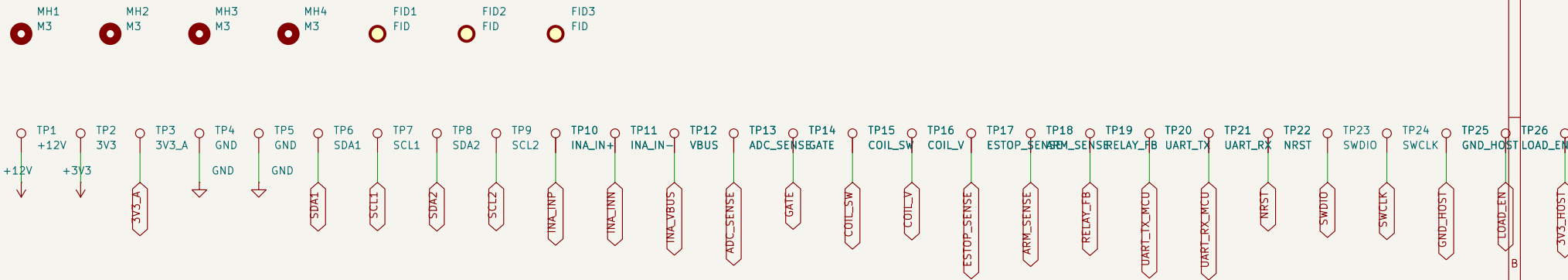
Rev: RC1

Id: 10/11

Test points, mounting, fiducials

OSBAMS Rev.2 Controller RC1 – REVIEW CANDIDATE, NOT FOR FABRICATION. Evidence: VERIFIED_LOCAL / USER_RELAYED_MANUFACTURER / UNVERIFIED (see evidence register).

Test points: +12V, 3V3, 3V3_A, GND x2, SDA1, SCL1, SDA2, SCL2, INA_IN+, INA_IN–, VBUS, ADC_SENSE, GATE, COIL_SW, COIL_V, ESTOP_SENSE, ARM_SENSE, RELAY_FB, UART_TX, UART_RX, NRST, SWDIO, SWCLK (+ LOAD_EN, GND_HOST, 3V3_HOST).



Manufacturer–document verification is a fabrication gate

RELEASE CANDIDATE 1 – NOT FOR FABRICATION

OSBAMS Rev.2 Controller

Sheet: /10_TestPoints_Mech/

File: 10_TestPoints_Mech.kicad_sch

Title: Test points, mounting, fiducials

Size: A4

Date: 2026–10–01

Rev: RC1

KiCad E.D.A. 10.0.6

Id: 11/11